



Why Language Models-Hallucinate

- Arnav Sinha

Why Language Models Hallucinate

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Abstract

Like students facing hard exam questions, large language models sometimes guess when uncertain, producing plausible yet incorrect statements instead of admitting uncertainty. Such “hallucinations” persist even in state-of-the-art systems and undermine trust. We argue that language models hallucinate because the training and evaluation procedures reward guessing over acknowledging uncertainty, and we analyze the statistical causes of hallucinations in the modern training pipeline. Hallucinations need not be mysterious—they originate simply as errors in binary classification. If incorrect statements cannot be distinguished from facts, then hallucinations in pretrained language models will arise through natural statistical pressures. We then argue that hallucinations persist due to the way most evaluations are graded—language models are optimized to be good test-takers, and guessing when uncertain improves test performance. This “epidemic” of penalizing uncertain responses can only be addressed through a socio-technical mitigation: modifying the scoring of existing benchmarks that are misaligned but dominate leaderboards, rather than introducing additional hallucination evaluations. This change may steer the field toward more trustworthy AI systems.

Kalai, A. T., Nachum, O., Vempala, S. S., & Zhang, E. (2025). Why language models hallucinate. arXiv preprint arXiv:2509.04664. [Link](#)

TECH · LAW

Humiliated lawyers fined \$5,000 for submitting ChatGPT hallucinations in court: 'I heard about this new site, which I falsely assumed was, like, a super search engine'

BY RACHEL SHIN
June 23, 2023 at 12:40 PM EDT



AI IN COURT

Misinformation Expert Shatters His Credibility

"[A] credentialed expert on the dangers of the siren call of relying too heavily on AI, no less."

EUGENE VOLOKH | 1.10.2025 7:10 PM



How Medical AI Can Fight Hallucinations by Staying Focused - Worth

Dec 11 - Marshall Honorof

Stanford HAI
Generating Medical Errors: GenAI and Erroneous Medical References

Feb 12

Kavita MD
Decoding AI hallucinations in health care: Embracing a new era of medical innovation

Dec 2

AI could help doctors stay up to date with diagnoses : Shots - Health News

Oct 25 - Craig LeMay

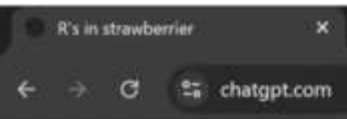
Courier
A Call to Address AI "Hallucinations" and How Healthcare Professionals Can Mitigate Their Risks

Sep 5, 2023

ChatGPT's 'hallucinations' make it difficult to trust for advice on cancer care

Aug 24, 2023 - Chad Van Allen

There are 3 letter "r"s in the word "strawberry."



How do you know?



Good question! I just looked at each letter in the word "strawberry" one by one and counted every time the letter "r" appeared.

Here's the breakdown:

- s
- t
- r (1)
- a
- w
- b
- e
- r (2)
- r (3)
- i
- e
- r

So, I counted three "r"s total. Want me to show you a different way to count letters?

Hallucination definition(s)

The screenshot shows the Merriam-Webster website with the search term 'hallucination'. The left sidebar contains links for 'Dictionary', 'Thesaurus', 'Synonyms', 'Synonym Chooser', and 'Example Sentences'. The main content area displays the definition of 'hallucination' as a noun, with a red box highlighting the third definition: 'computing : a plausible but false or misleading response generated by an artificial intelligence algorithm'. Below this, a quote from Prabhakar Raghavan is visible.

Merriam-Webster

Dictionary Thesaurus

hallucination

Games Word of the Day Grammar Word Finder

Dictionary

Definition

Synonyms

Synonym Chooser

Example Sentences

3 **computing** : a plausible but false or misleading response generated by an artificial intelligence algorithm

"This type of **artificial intelligence** we're talking about can sometimes lead to something we call *hallucination*," said Prabhakar Raghavan in an interview with Germany's Welt am Sonntag newspaper published on Saturday. "This is then expressed in such a way that a machine delivers a convincing but completely fictitious answer."

A hallucination is when a language model produces information that is factually incorrect, invented, or unsupported by its inputs. Hallucinations arise from the model's statistical guessing rather than grounded knowledge

Big Picture

The paper's central claim is:

***Hallucinations are not accidental glitches.
They follow directly from the mathematics of how modern LLMs work.***

Two major phases of the LLM pipeline:

1. **Pretraining** - The phase where the model learns language from massive datasets via next-token prediction
2. **Post-training** - The phase where humans, or other models, teach it to be helpful, polite, harmless, etc.

The authors argue:

- A. Pretraining produces models that *must* hallucinate because no model can flawlessly generate valid text in all cases.
- B. Post-training **encourages** hallucinations by creating **test-taking incentives** (guessing when unsure gives higher scores).

How Pretraining Works

Main Task: Predict the next token given all previous tokens.

The pretraining objective is:

$$\min_{\theta} \mathbb{E}_{x \sim \text{training}} [-\log p_{\theta}(x)]$$

- For every piece of text in the training corpus,
- The model is rewarded for assigning higher probability to the next token the text actually contains.

Pretraining teaches the model to imitate the statistical distribution of text.

It does *not* teach the model what is true.

It teaches what humans *tend to write*.



LLMs learn **language distributions**, not **truth distributions**.

Why Is This Important

Since the model is learning from text:

- Human text contains **errors, typos, misconceptions, contradictions**, etc.
- Some facts appear many times (“Einstein born in 1879”)
- Some facts appear **once** (e.g., an obscure person’s birthday in one obituary)
- Some facts are never stated (rare birthdays, personal facts, small-town history)

*Thus, the model must **infer, generalize, and guess.***

Even if the data were magically perfect (no errors), pretraining still forces the model to:

- Estimate probabilities from limited samples
- Generalize beyond what it has seen
- Use inductive biases that can misfire
- Compress enormous complexity into a finite number of parameters

These are the “statistical pressures.”

Why hallucinations are unavoidable even with perfect training data

Imagine a very simple task:

Given a string, tell me whether it is

VALID (true, correct) $\rightarrow +1$

INVALID (false, hallucination) $\rightarrow -1$

If we design any classifier to do this over the space of *all* possible text strings:

- The classifier is imperfect.
- It must misclassify some strings.
- No finite dataset can let it perfectly distinguish *all* valid strings from *all* invalid ones.

No classifier is perfect unless it sees infinite data or has infinite capacity.

Generation is harder than classification

Here is the paper's core insight:

**When a model generates text, it implicitly performs a classification:
“Is this token (or sentence) valid or invalid?”**

Every time the LM chooses a next token:

- It must implicitly decide which completions are acceptable.
- If it misclassifies, it chooses an invalid token → **hallucination**.
- Thus:

$$\text{Hallucination rate} \geq 2 \times \text{classification error rate}$$

This is a major theorem in the paper.

Some facts cannot be learned from data, no matter how good the model is

Example: birthdays.

Common facts (seen many times) are learned well; rare facts (seen once) are unreliable.

The authors use Good-Turing missing mass theory to quantify this: ***The probability that a new sample will fall into an “unseen” category is approximately the fraction of samples seen exactly once.***

Thus, if 20% of facts appear once:

→ the model should be expected to **hallucinate** *at least* ~20% of similar facts.

This is not a bug.

This is **statistics**.

Even perfect data cannot overcome limits of representation

Even if:

- all data is correct,
 - contains no contradictions,
 - includes every fact multiple times,
- the model still faces:

Finite capacity

LLMs compress massive distributions into a finite parameter space.

Compression causes approximation error → mistake → hallucination.

Complex decision boundaries

Some classification boundaries cannot be represented perfectly by the architecture. E.g., linear thresholds for highly nonlinear functions.

Non-learnable patterns

For many tasks, there is no learnable pattern.

Example:

"There are 364 wrong birthdays for every correct one."

How Post-Training Reinforces (Not Reduces) Hallucinations

After pretraining, models go through **post-training** methods:

- **RLHF** — Reinforcement learning from human feedback
- **DPO** — Direct Preference Optimization
- **RLAIF** — AI feedback instead of humans

These methods accidentally encourage hallucinations by imitating how humans score answers.

Why?

1. Most benchmarks use binary scoring -> ***Expected score(guess) ≥ Expected score(IDK)***
2. During RLHF:

The model learns to **never say “I don’t know”** | The model learns to **guess confidently** | The model becomes a **test-taking machine**, not a truth-telling machine

3. Models are punished for *honesty*. Models that admit uncertainty score lower than models that hallucinate confidently.

Author's Proposed Solutions

- Add explicit confidence thresholds (e.g., 50%, 75%, 90%) to every benchmark question.
- Use negative marking so wrong answers lose points and “I don’t know” scores zero.
- Modify mainstream evaluations like MMLU and GPQA so they reward abstention when uncertain.
- Put the confidence requirement directly in the prompt for every question.
- Evaluate models based on **behavioral calibration**: ensure the model behaves according to the confidence threshold, meaning it should answer only when its internal confidence exceeds the target and explicitly abstain (“I don’t know”) when below it.

Behavioral calibration focuses on the correctness of the model’s decision-making behavior (answer versus abstain) rather than on probability estimates or hedging language. A model is behaviorally calibrated when it consistently answers only when justified by its confidence and withholds answers when unsure.

**THANK
YOU**